

Nitish Rathore

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EDUCATION

PES University

Dec. 2020 – May 2024

B.Tech in Computer Science & Engineering (Specialization: Machine Intelligence and Data Science)

Bengaluru, India

- CGPA: 8.66/10
- Coursework: Machine Intelligence, Data Structures & Algorithms, Software Engineering, Object Oriented Programming in Java, Database Management Systems, Operating Systems, Cloud Computing, Big Data

PROFESSIONAL EXPERIENCE

KangaSys

Bengaluru, India

Software Engineer

March 2025 – Present

- Architected multi-tenant SaaS IoT analytics platform processing 10M+ daily sensor events from Honeywell Building Management Systems using ASP.NET Core microservices, ClickHouse time-series database, PostgreSQL, and AWS serverless infrastructure for real-time monitoring and analytics.
- Engineered analytics pipelines delivering 100–500ms query latency on million+ record datasets via optimized ClickHouse aggregation, indexing strategies, and query optimization, enabling real-time dashboards and alerting systems.
- Developed Next.js 14 dashboards with 30+ customizable widgets, drag-and-drop layouts, Chart.js visualizations, and AWS Cognito MFA authentication with RBAC ensuring secure multi-tenant data isolation.
- Implemented serverless reporting (AWS Lambda + SQS) for async PDF/CSV exports, tenant-specific RBAC with Permit.io, and automated CI/CD with Docker and Bitbucket deploying to AWS ECS.

Pronisi,Inc.

Bengaluru, India

Software Engineer

Jan. 2024 – Feb. 2025

AI/ML Intern

June 2023 – Dec. 2023

- Led Computer Vision and Pattern Recognition team developing SlamDunk.AI, AI-powered app for analyzing users' sports performance and providing personalized feedback for improvement, managing full lifecycle of deep learning models across various applications using PyTorch, TensorFlow, Keras, and AWS.
- Utilized Detectron2, YOLOv7, BlazePose, Mask R-CNN, Fast R-CNN, ResNet, and MobileNet models for extracting key data from user videos for recommendations and sports analysis.
- Managed end-to-end data pipeline including data collection, cleaning, annotation, and preprocessing for multi-sport video datasets supporting tennis, cricket, basketball, and golf analysis.
- Developed solutions for fast-moving ball detection, human pose detection, and key point estimation for sports equipment, achieving over 80% accuracy in critical models as part of Machine Learning R&D.
- Productionized CV inference pipelines with ONNX optimization (40% faster inference), GPU-enabled Docker with NVIDIA CUDA, deployed scalable microservices on AWS ECS with CloudWatch monitoring.

International Institute of Information Technology (IIIT) Bangalore

Bengaluru, India

Research Intern

June 2022 – Aug. 2022

- Developed CNN-based classifiers for automated LULC mapping from multispectral satellite imagery, implementing preprocessing pipelines (radiometric calibration, noise reduction, cloud masking) for pixel-level classification of urban, agricultural, forest, and water bodies.
- Designed band selection techniques using statistical analysis to identify optimal spectral bands, reducing computational complexity while maintaining classification accuracy across diverse land cover types.
- Utilized GRASS GIS and Python libraries (GDAL, Rasterio, GeoPandas) for spatial processing and accuracy assessment, contributing to research achieving ~90% classification accuracy on benchmark datasets.

ACADEMIC PROJECTS AND PUBLICATIONS

Recommending Solutions for Bangalore: Leveraging Intelligent Systems with CA

<https://bit.ly/48C5ef5>

- Developed intelligent urban planning system combining CA-GA urban growth predictions with time series analysis for population forecasting until 2030, achieving 2.98% MAPE accuracy using linear regression on WorldPop datasets.

- Identified congestion-prone corridors by analyzing routing discrepancies across Google Maps, OSRM, and OSMnx, recommending infrastructure interventions including flyovers, road widening, and public transport enhancements where routes non-coincide.
- Predicted water infrastructure requirements using Hazen-Williams hydraulic equations and volumetric flow calculations, forecasting 2030 water demand based on CA-GA urban expansion and recommending pipeline diameter upgrades aligned with BWSSB standards.
- Generated optimized waterway connectivity networks as GeoTIFF files incorporating land use patterns, pipeline capacity, and population projections for sustainable water resource management.
- Estimated future electricity demand by analyzing correlation between urban sprawl patterns, land use changes, transportation growth, and green technology adoption.

Urban Growth Modelling using Cellular Automata and Genetic Algorithm

(<https://bit.ly/3KUuow6>)

- Developed integrated CA-GA model for spatiotemporal urban growth prediction in Bangalore (2017–2025), calibrating CA transition rules using GA with SSIM as fitness function, achieving SSIM values of 0.83–0.84 validated against Landsat satellite imagery from USGS Earth Explorer.
- Processed and integrated multi-source geospatial datasets using QGIS: OpenStreetMap data (parks, schools, metro routes, road networks, waterways), Landsat imagery with radiometric and atmospheric corrections, and population density data, converting raw imagery into detailed LULC maps through advanced classification techniques.
- Designed CA model incorporating proximity factors (metro, roads, universities, central business district, water sewage lines, population density) and additional input maps (drinking water, electricity) to enhance prediction accuracy, with GA optimizing neighborhood effects and conversion probabilities through iterative parameter calibration.
- Validated model against 2017–2022 satellite-based LULC maps, quantifying 90.04 sq. km urban expansion, 25.88 sq. km vegetation loss, 7.19 sq. km water body reduction, and 58 sq. km barren land reduction, demonstrating accurate prediction of compact growth and improved urban sprawl capture through GA integration.

Yet Another Kafka: Distributed Messaging System

(<https://bit.ly/49eNFkn>)

- Implemented Kafka-like distributed pub-sub messaging system in Python demonstrating leader election, replication, fault tolerance, and scalable message routing with Apache ZooKeeper for coordination and Redis for persistent metadata storage.
- Developed dual implementations: socket-based using raw TCP for low-level optimization, and RabbitMQ-based for rapid prototyping, both supporting dynamic topic partitioning with three-fold active replication across all brokers.
- Implemented fault-tolerant architecture with Raft-inspired leader election, heartbeat monitoring, and automatic failover redistributing topic leadership when broker failures detected, ensuring continuous operation with single broker availability.
- Built producer/consumer APIs supporting sync/async publishing, consumer group management with load-balanced partition assignment, offset tracking for delivery semantics, and rebalancing protocols for membership changes.

Microservice Communication with RabbitMQ

(<https://bit.ly/457wShz>)

- Architected distributed microservices system with Flask-based producer service exposing REST APIs (health check, CRUD operations) and four specialized consumer services handling asynchronous message processing, implementing pub-sub and work queue patterns using RabbitMQ direct exchanges and durable queues.
- Integrated MongoDB database operations (insert, delete, read) across consumer services with message acknowledgment mechanisms ensuring reliable data persistence, and implemented health check endpoints for monitoring distributed system components.
- Dockerized complete application stack using docker-compose orchestrating producer, four consumer services, MongoDB database, and RabbitMQ broker across custom Docker networks, enabling scalable and reproducible microservices deployment.

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, TypeScript, SQL, C/C++

Frameworks/Tools: PyTorch, TensorFlow, ASP.NET Core, Next.js, React, ClickHouse, Redis, Kafka, Docker, Git

Infrastructure & Platforms: AWS (Lambda, SQS, Cognito, EC2, S3), Linux, Windows

ACHIEVEMENTS

- Won 1st Place in Arthemina Hackathon for deep fake detection project
- Best Presenter Award - IEEE BDAI 2023 International Conference
- Second Runner Up at Terrathon Ideathon
- 3X MRD Scholar (Merit-based Scholarship)